

TECHNICAL UNIVERSITY OF DENMARK



34840 Fundamentals of Acoustics and Noise Control

LAB EXERCISE

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1 Introduction

This report documents the procedures and findings of an experiment regarding room acoustics carried out on 22nd of October 2025 at building 355. The primary purpose of this exercise was to practically investigate and quantify three fundamental acoustic characteristics of a standard rectangular room: room modes, reverberation time (T_{rev}), and sound power level (L_W).

Understanding these characteristics is useful for practical noise control and room acoustics applications, because they describe the sound quality, speech intelligibility, and sound pressure distribution within enclosed spaces. The experiment was conducted in a dedicated test room with approximate dimensions of 4.38 m (length), 3.13 m (width), and 3.29 m (height). This report consists of three main objectives:

- **Modal Analysis:** To identify the five lowest room eigenfrequencies (modes) through measurement and compare these values with theoretical predictions derived from the room dimensions. This section also includes identifying the pressure distribution in the space of these modes.
- **Reverberation Time Measurement:** To measure the reverberation time across octave bands in four distinct room configurations: an empty state, with added sound absorbers, with added diffusers (scatterers), and with diffusers only.
- **Sound Power Estimation:** To estimate the A-weighted sound power level of a vacuum cleaner using the measured average sound pressure level and reverberation time, based on the principles of the reverberation chamber method.

The following sections detail the instrumentation and procedures, present the experiment results, and give insight into how acoustic measurements are conducted in real-world scenarios. Although the experiment procedure does not exactly follow the guidelines outlined in ISO 3382-2 standard "Measurement of room acoustic parameters - Part 2: Reverberation time in ordinary rooms" [1], many of the steps necessary in the standard have been taken here.

2 Relevant Theory

This section briefly summarises the relevant theory; detailed explanations and derivations can be found in [2], [3], [4].

2.1 Room Modes

Room modes, or eigenfrequencies, are the natural patterns formed by standing waves that occur when sound waves reflect between parallel room surfaces. The eigenfrequencies (f_{n_x, n_y, n_z}) in a rigid-walled rectangular room are calculated using the following equation

$$f_{n_x, n_y, n_z} = \frac{c}{2} \sqrt{\left(\frac{n_x}{L_x}\right)^2 + \left(\frac{n_y}{L_y}\right)^2 + \left(\frac{n_z}{L_z}\right)^2} \quad (1)$$

where c is the speed of sound in air $c \approx 343$ [m/s], L_x, L_y, L_z are the room dimensions, and n_x, n_y, n_z are integers representing the mode orders in the respective directions. The modes present in the room can be classified as axial modes, where one index is non-zero (e.g., $n_x, 0, 0$) such that sound travels between two parallel surfaces, tangential mode, where two indices are non-zero (e.g., $n_x, n_y, 0$) such that sound travels around four surfaces, and oblique mode, where all three indices are non-zero (e.g., n_x, n_y, n_z) such that sound involves all six surfaces.

2.2 Reverberation Time T_{rev}

Reverberation time is defined as the time it takes for the sound pressure level to decay by 60 [dB] after the sound source is abruptly stopped. Due to potential background noise limitations, a full 60 [dB] decay is

rarely measured. Instead, the decay rates T_{20} , measured over a 20 [dB] decay range (typically from -5 [dB] to -25 [dB]) and multiplied by 3 to estimate T_{60} , and T_{30} measured over 30 [dB] decay range (typically from -5 [dB] to -35 [dB]) and multiplied by 2 to estimate T_{60} are used.

In this context, absorbers like the porous panels used in this experiment reduce the sound energy and lower the reverberation time. Meanwhile, diffusers scatter incident sound energy in many directions to create a more diffuse sound field, which can influence T_{rev} by directing energy toward existing absorbers.

2.3 Sound Power Level (L_W)

Sound power (W) is the total acoustic energy radiated by a source per unit time, and the sound power level (L_W) is the energy measure of that source. In a reverberation room, the sound power level L_W can be estimated as follows

$$L_W = L_p - 14 \text{ dB} + 10 \log(V/V_0) - 10 \log(T_{rev}/t_0) \quad (2)$$

where V_0 is the reference volume $V_0 = 1[m^3]$, t_0 is the reference time $t_0 = 1[s]$ and L_p is the spatially averaged sound pressure level. Lastly, A-weighting is applied using the values in Table 1.3.2 in [2] to approximate the human ear's response to sound.

3 Method

To address the aforementioned objectives, the equipment and procedure outlined in [5] were used. The upcoming section details the choices and calculations made throughout the experiment.

3.1 Identifying room resonances

Before the lab experiment, the lowest five room modes for the given room dimensions were calculated using Equation 1 and are presented in table 1. During the exercise, the modes displayed in table 2 were investigated. The discrepancy between the values in the two tables is addressed in the section section 4.

Mode ID	Frequency
(1,0,0)	39.2 Hz
(0,0,1)	52.1 Hz
(0,1,0)	54.8 Hz
(1,0,1)	65.2 Hz
(1,1,0)	67.3 Hz

Table 1: *First five calculated room modes. $x = 4.38m$, $y = 3.13m$, $z = 3.29m$.*

Mode ID	Frequency
(1,0,0)	39.2 Hz
(0,1,0)	54.8 Hz
(1,1,0)	67.3 Hz
(2,0,0)	78.3 Hz
(2,1,0)	95.6 Hz

Table 2: *Room modes investigated during the experiment.*

3.2 Investigating reverberation time

The room measurement method loosely followed the standard for measuring room acoustics, in which a random noise signal was used to excite the room. However, instead of using an omnidirectional sound source, the procedure employed a directional speaker placed in the corner of the room. This approach is still valid because it maximizes modal excitation and energizes all low-frequency room modes. The corner is the pressure anti-node for every mode, therefore forcing the sound to interact immediately with three surfaces by placing the speaker in the corner helps to build a more diffuse sound field.

The measurements used the interrupted noise method, where a random noise signal is used to excite the room until it reaches the steady state, and then is stopped abruptly to record the sound decay. Alternatively,

more complex methods like the Integrated Impulse Response¹ (Schroeder's Method) or Cross-Correlation² using signals like sine sweeps [6] could be used. However, the interrupted noise method offers a direct observation of the decay characteristics of the excited room modes. The results were also averaged over five runs for each microphone position, and the background noise was also measured to determine the signal-to-noise ratio (SNR) of the results. It is also important to note that no one was located in the room during the experiment to prevent absorption by the human body that would skew the results.

In addition to averaging over five measurements, the microphone used to measure the sound was positioned in three different arrangements shown in Figure 1 & Figure 2. These positions were chosen to evaluate the differences between the sonic conditions at various places in the room.

- **Microphone position 1** does not directly correspond to the nodes or antinodes of any of the fundamental room modes. The idea was that it would provide a result that is less affected by the room modes in the low frequencies.
- **Microphone position 2** is deliberately placed close to a wall and a corner to achieve a result that captures room mode amplification.
- **Microphone position 3** is also deliberately placed in the centre of the room to achieve a result that omits the fundamental room modes and the odd-numbered multiples of them.

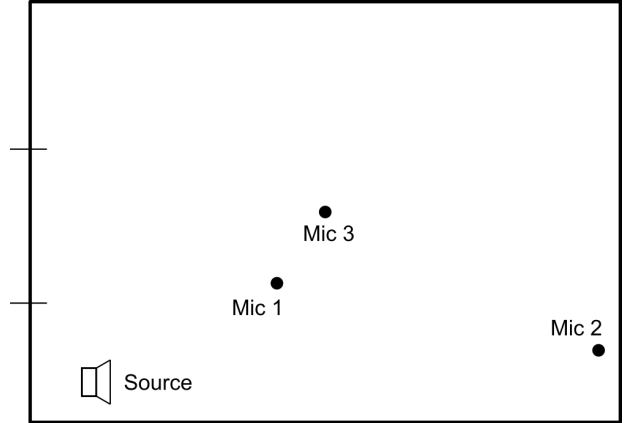


Figure 1: *Room measurement microphone and sound source conceptual locations in the room.*

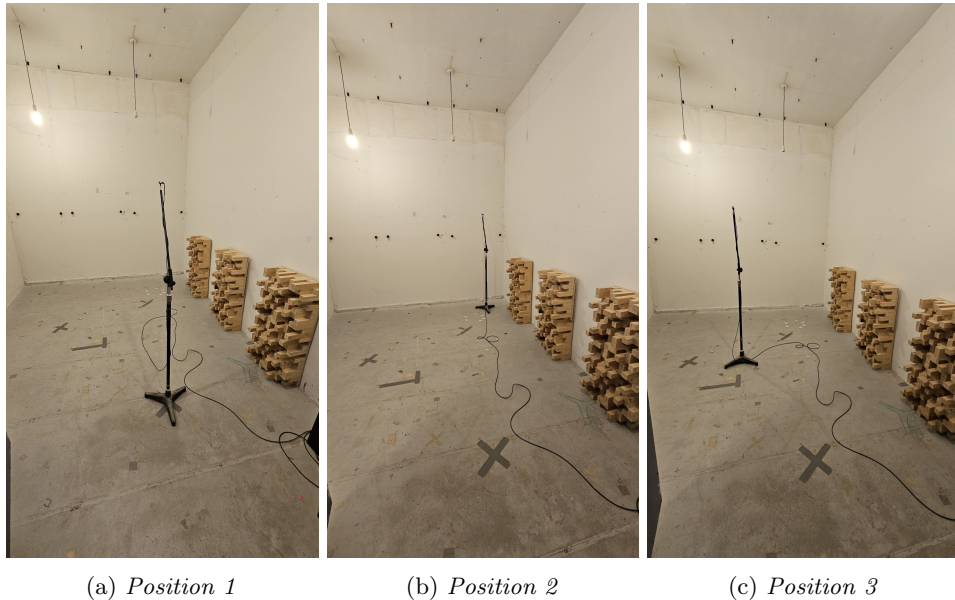


Figure 2: *Microphone positions in reverberation chamber.*

¹Calculates reverberation time by integrating the impulse response to provide a smooth decay-rate curve

²(For instance, Maximum Length Sequences) estimates the decay rate by correlating the measurement signal with the known excitation signal, so that it does not significantly suffer from noise.

During the exercise, four separate measurement conditions were explored, including an empty room, either with porous absorbing material or with diffusers, and with both. The arrangements used in the experiment are shown in Figure 3, where the absorbers could only be placed on the wall opposite the speaker, while the diffusers were chosen to be placed on the wall adjacent to it.

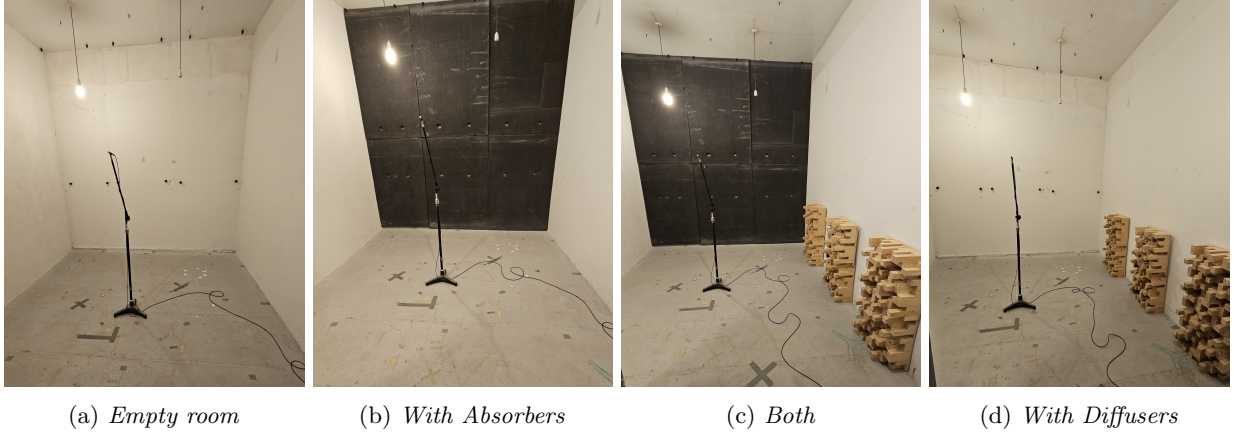


Figure 3: *All measurement conditions*

It was chosen to place diffusers along one of the bare surfaces to, in theory, reduce the specular reflections between the two untreated walls. In practical applications, it is imperative not to leave two parallel planes without either absorption or diffusion. Otherwise, an audible flutter echo would be present between them. Moreover, in this case, the area of the diffuser is possibly too small to provide sufficient scattering. Furthermore, the ceiling and the floor remain bare, which would lead to an echo between them.

These diffusers are similar to the **Q**uadratic **R**esidue **D**iffuser (QRD) type. What's more, they are 2D, which means they provide scattering of the incident wave in two planes. A classical QRD is based on a prime number, whereas these diffusers have 12 elements per side.

Approximating the characteristics of this diffuser to a 2D QRD panel with prime 11 as its base, we get the following:

$$D = n^2 \bmod p \quad (3)$$

where D = desired depth of the panel, n = integer and p = chosen prime number [7].

Eq. 3 provides the maximum depth (or height in this case) of the individual elements of a QRD panel. This depth as well as the width of the elements, determines the frequency range at which the panel is effective. The overall depth of the panel is around 20 [cm], and a single element's width is 4 [cm]. Using

$$f_l = \frac{c}{2 \cdot 0.2} = \frac{343}{0.4} \approx 858 \text{ Hz} \quad (4)$$

as a simple approach to calculating the lower diffusion limit, and

$$f_u = \frac{c}{2 \cdot 0.04} = \frac{343}{0.08} \approx 4.29 \text{ kHz} \quad (5)$$

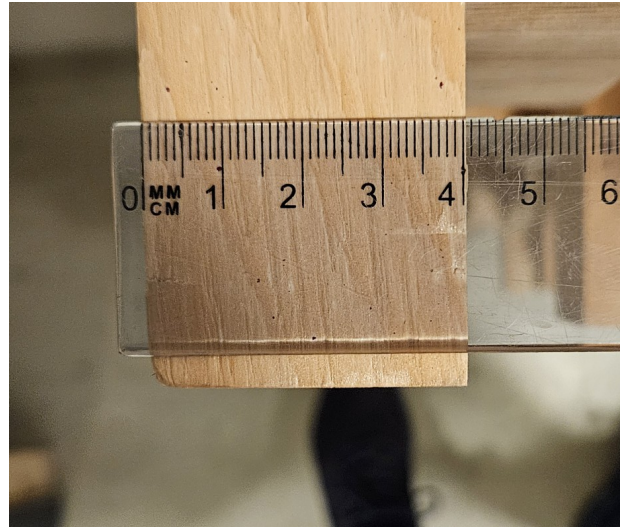


Figure 4: *Diffuser element width.*

to calculate the upper limit (both stemming from the idea that to affect a sound of certain frequency, an object with dimensions of at least half a wavelength is necessary), it is evident that these diffusers have little to no effect on the lower frequencies in the room and only affect the high frequencies by creating a scattered sound field, it being more akin to a diffuse sound field, which provides a more natural-sounding environment in a room [4].

The placement of the absorbers was not variable during the experiment, but covering an entire surface of a room with absorbing material is a common way how acoustic treatment is done. Usually it is the ceiling that is treated, but in order to reduce reverberation time in a room equivalent absorption area is important. One advantage to drop-in ceiling construction is the added benefit of the ceiling panels acting as a membrane for low frequency absorption. This is often impractical for wall paneling. Leaving a cavity behind the paneling increases low frequency absorption at the cost of reducing the size of the room. Furthermore, the acoustic panels used in the experiment are no more than 5 [cm] thick and therefore provide little effect in absorbing the low frequency sound energy. The absorption data for 50 [mm] thick polyurethane panel is shown in table 3³.

Table 3: *Flexible 50 [mm] thick polyurethane foam panel absorption coefficients in octave bands.*

Frequency bands	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz
α	0.25	0.5	0.85	0.95	0.90	0.90

3.3 Sound Power of a vacuum hover

To investigate the sound power of a vacuum cleaner, it was placed in the middle of the room, and the measurements were averaged over the three aforementioned microphone locations.

Placing the vacuum cleaner centrally ensures the microphone measurements are taken in the diffusive sound field of the room. By averaging the sound pressure level at multiple points, this method minimizes the effects of local standing waves and distance variations. This averaged result is then directly used to calculate the sound power level (L_W) using Equation 2 of the source, which is independent of the test room’s specific acoustics.



Figure 5: *Sound power measurement of a vacuum cleaner at microphone position 1.*

4 Results & Discussion

4.1 Room modes

Room mode investigation led to a greater appreciation of the nature of eigenfrequencies in a given room and the potential pitfalls of not accounting for them. From the loudest sections in the room, such as the corners, to the quietest, an audible difference was noticeable and was confirmed by a decibel meter phone app showing the change by roughly 20 [dB]. Even though the calculated room modes did not always align with the actual loudest resonances and deviated from the theoretical calculations by ± 1 [Hz] based on hearing perception, it can be said that this falls within the perception error, hence making the theory very applicable. This aligns well with the reverberation results of the empty room condition, as they were very high due to the rigid walls.

³UK’s Institute of Acoustics dataset.

4.2 Reverberation time

The reverberation time of the room was determined by driving pink noise in the room, recording five decay curves in each of the three microphone positions, and averaging them per octave band, from 125 to 4000 [Hz].

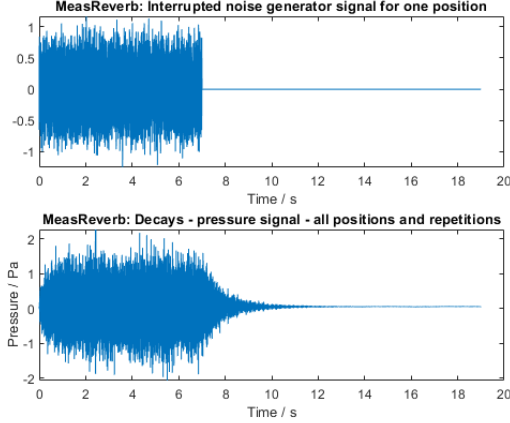


Figure 6: *Single measurement of pressure signal after noise generator is suddenly turned off, for an empty room, test run.*

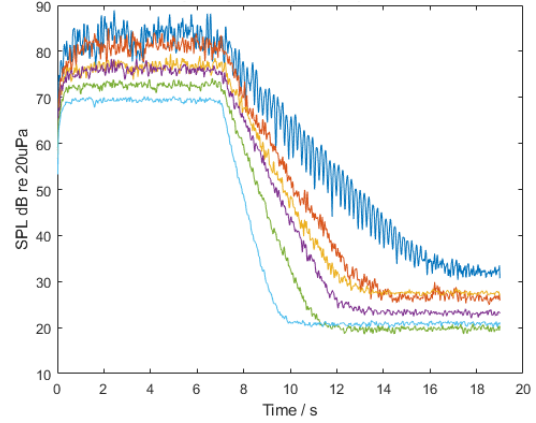


Figure 7: *Sound pressure level in dB for one microphone position in an empty room.*

Figure 6 and Figure 7 show the results of the reference measurement for the empty room, and recording only in one position and with a single averaging. Figure 6 shows the pink noise signal with a steady-state duration of 7 seconds, and the recorded decay of the noise in the room. Figure 7 shows the decay of the noise by octave band, exhibiting a higher SPL as the frequency decreases, since pink noise has a power spectral density inversely proportional to the frequency, as well as a longer decay time at low frequencies, which is consistent with the expected behavior of the low frequencies in a room with rigid walls.

Figure 8 shows that the averaged measurements of the empty room follow the same overall decay trend as the reference data across most octave bands. However, the averaged measurements reach a significantly higher noise floor, which limits the observable decay and interferes with a direct comparison in the late portion of the curve. The elevated noise floor is likely caused by faults in the measurement equipment and additional background noise produced by ongoing construction work during the measurement session.

If both results had the same noise floor, it would be expected that the averaged measurements would continue to decay further and converge more closely to the reference curves. Since averaging over several microphone positions reduces spatial variability, the averaged measurements would likely show a smoother decay at the end of noise and, in some bands, even more reliable T_{rev} values than the single-position single-average reference. In particular, the decay slopes in the early 20–30 dB range have a strong match between the reference and the measured data, suggesting that the primary discrepancy comes from the measurement noise rather than from acoustical differences. Under ideal noise conditions, both curves should reach comparable minimum levels and exhibit nearly identical reverberation times, especially at mid and high frequencies where the sound field is more diffuse.

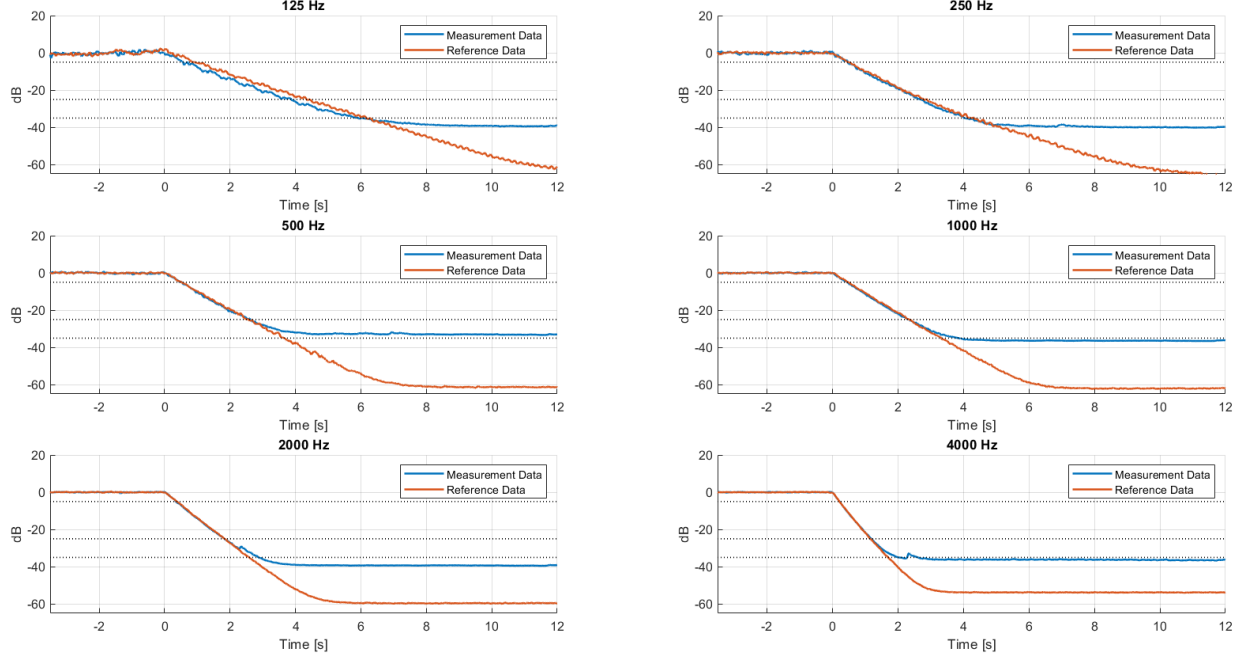


Figure 8: Comparison of reference data provided of a single position measurement in an empty room to averaged measurements over three positions for an empty room.

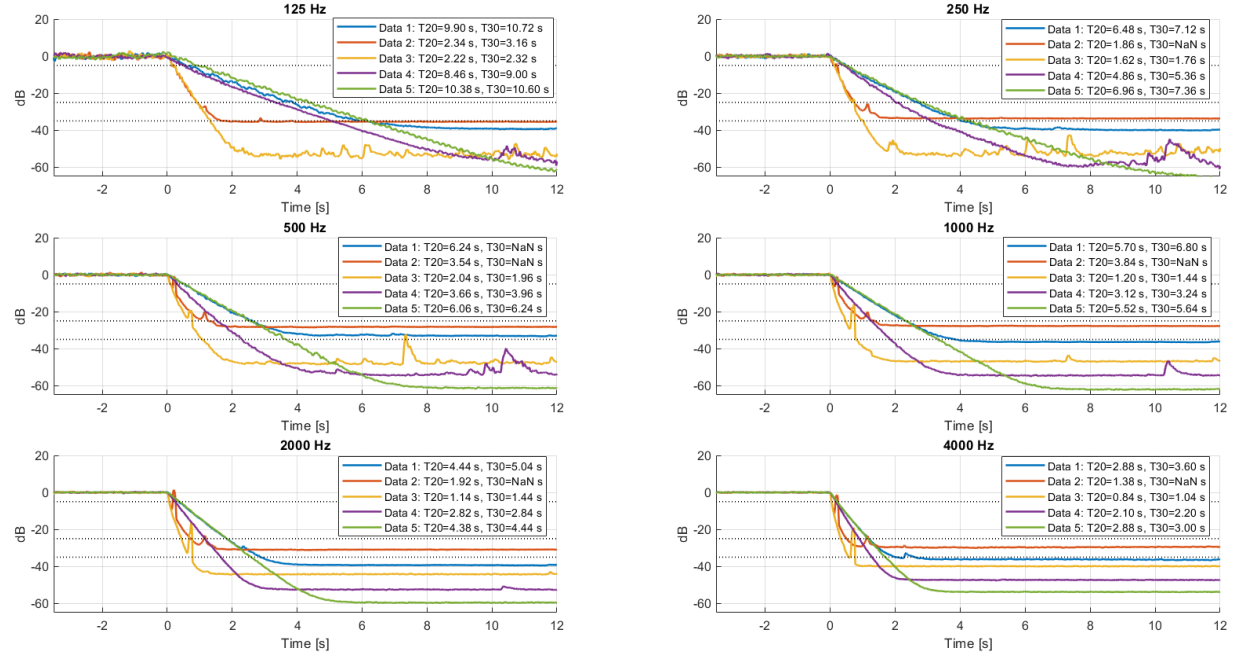


Figure 9: Decay curves including reverberation times T_{20} and T_{30} of the six octave bands. Data 1: empty room, Data 2: room with absorbers, Data 3: room with absorbers and diffusers, Data 4: room with diffusers only, and Data 5: reference data of a single empty room measurement.

Figure 9 shows the results of the measurements for the four room conditions. It can be seen that for the first and second measurements (Data 1 and Data 2), the same problems with noise floor were encountered, causing Data 2 to plateau before reaching T_{30} in all octave bands except the 125 [Hz] band.

Regardless, it is possible to compare the T_{20} values between the different measurement conditions. It is clearly evident that with the absorbers present in the room, the reverberation time decreases. This was expected. The difference between Data 2 and Data 3 was the diffusion panels. It can be seen that the condition with absorption panels and the diffusers provides significantly lower reverberation time in 500 Hz, 1000 Hz and 2000 Hz octave bands. This can be attributed to the reduction of specular reflections between the two bare walls, the ceiling and the floor. Diffusers provide scattering in their effective frequency range, ensuring that the sound energy is reflected in various directions, including towards the absorption panels.

Evaluating the difference between Data 1 and Data 4, i.e., without and with diffusion panels, the results show a reduction in reverberation time. This can be attributed to increased scattering in the room, reduced specular reflections, and less flutter echo.

4.3 Sound Power

The Sound Power Level (SWL) was determined through the *free-field method* in the reverberation room by averaging the mean squared pressure of the three microphone positions, and using an approximation of the reverberation time based on the measured value of T_{30} ($T_{rev} = 2 \cdot T_{30}$). Then, the respective correction factor of the A-weight filter was applied to the SWL of each octave band.

In this case, it was decided to use T_{30} to estimate T_{rev} because it includes more samples from the decay curve in its calculation, which improves the stability of the decay slope and reduces sensitivity to small fluctuations in measurements. In addition, it requires less extrapolation, reducing the effect that small deviations may have on the T_{rev} estimate.

Since the SWL is a characteristic of the source, it should not be affected by the room acoustics. A diffuse field is needed, with equal probability that sound comes from any direction and with equal energy density at all points in the room [2]. That is why these types of measurements are conducted in a reverberation room, since its characteristics allow us to assume that its sound field is diffuse. For this reason, it was decided to leave the diffusers in place, in order to create a more uniform and diffuse sound field.

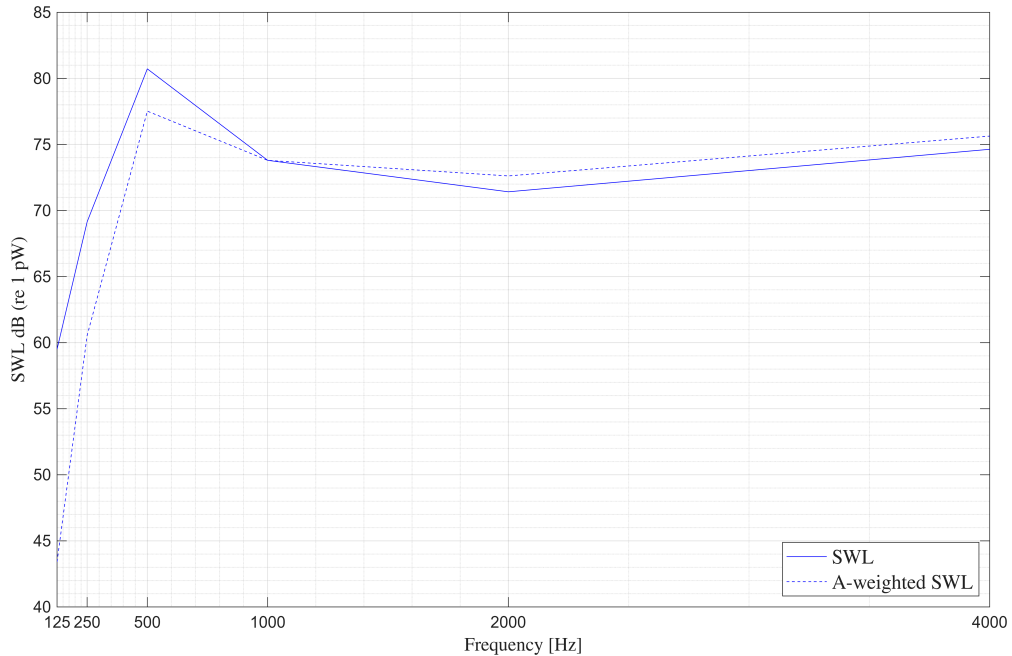


Figure 10: *Averaged Sound Power Level of the three microphone positions, unweighted and A-weighted, with a vacuum cleaner turned on inside the room.*

In Figure 10, it is observed that the 500 [Hz] octave band was the one with the maximum SWL, both unweighted and A-weighted, reaching 80.7 [dB] and 77.5 [dB(A)]. For the octave bands below 500 [Hz], the SWL and A-weighted SWL drop significantly due to the reduced spectral components in low frequencies

of the sound output from the vacuum cleaner and the bigger attenuation factor of the A-weighting filter towards low frequencies. For the 1000 [Hz] octave bands, SWL and A-weighted SWL decrease, reaching the same level as the correction factor for this band is 0 [dB]. Both SWL have a slight decrease in 2000 [Hz] and increase in 4000 [Hz], but in these two octave bands, A-weighted SWL is greater since the correction factor is an amplification for these frequencies.

To calculate the combined broadband A-weighted SWL of the source, the summation of octave bands was used (replacing the SPL values for the SWL ones). The result was 81.35 [dB(A)] SWL.

Figure 10 shows that the vacuum cleaner concentrates more energy in the mid-frequencies range and has less energy in the low ones, meaning that the broadband noise of the vacuum cleaner generates more acoustic radiation in the mid-frequencies and not that much in the lower frequencies. However, it should also be considered that the sound power level is influenced by the reverberation time (Equation 2). Since low frequencies have lower absorption, this increases their reverberation time which consequently increases the value of this term and, therefore, reduces the estimated SWL. Another aspect that should be considered for the low SWL at low frequencies is the room modes. Since there are standing waves present in the reverberation room, which create zones of high and low sound pressure, it impacts negatively on the diffuse sound field, creating uneven spatial energy distribution at low frequencies.

5 Conclusion

Though the experiment yielded the relevant data, it was marred by a poor signal-to-noise ratio, which led to insufficient usable dynamic range to evaluate all parameters. The cause for these issues is still not clear. It is believed that equipment malfunctions, as well as the ongoing construction, could be the cause of them.

Despite the issues in the results, certain conclusions can be drawn. For instance, acoustic materials and diffusion elements have a great effect on reverberation time in a room. Also, room modes play a significant role in characterizing the sound profile of a room.

Sound power measurements can also be influenced by the room modes, the microphone positions, and the source position. Under the given conditions, it was possible to determine the sound power of a vacuum cleaner due to the reverberant nature of the test chamber. However, for better results, a room with more diffuse sound field would be beneficial.

References

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